Alzheimer's Disease Prediction Using Machine Learning Techniques

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***Abstract*—This work utilizes machine learning and explainable artificial intelligence (XAI) techniques to predict Alzheimer's Disease (AD) using structured clinical information. These include the OASIS-3 data involving MMSE, CDRSUM, CDRTOT, and demographics that were utilized to train and compare Logistic Regression, Random Forest, and Support Vector Machine models. Mutual Information and Recursive Feature Elimination (RFE) were used for the process of feature selection. SHAP (SHapley Additive exPlanations) was utilized to provide explanations to model predictions. In all the classifiers, Random Forest performed the most balanced trade-off and interpretability. The end-to-end pipeline provides a transparent, scalable, and cost-effective early AD prediction tool for assisting in the diagnosis.**

Introduction and Statement of the Problem

Alzheimer's Disease is a chronic and progressive neurological condition that affects cognition and memories and also has a significant impact on behavior. It is still the largest cause of dementia and affects more than 6 million Americans. Existing diagnostic methods usually happen in advanced stages and use costly or invasive procedures such as PET scans and analysis of cerebrospinal fluid. These approaches are not scalable or cost-effective despite their accuracy in the clinic. There is a pressing need for early, affordable, and interpretable diagnostic tools. This project explores using machine learning models on cognitive and clinical data to create an early AD diagnosis system that is both explainable and accurate for real-life applications.

Review of Literature

[**?**,]]sarraf2016 Shown to detect Alzheimer's very accurately using deep learning models based on fMRI data. [**?**,]]marcus2010 presented the Open Access Series of Imaging Studies (OASIS), which has precious longitudinal data to study the diagnosis of neurodegenerative diseases. [**?**,]]westman2012 showed that the combination of MRI and CSF markers enhanced AD performance in classification. While the employment of high-end images puts a limit on the opportunity to provide scalable results to a large population, other publications using general-purpose neuropsychology and clinical data stress that more uncomplicated non-imaging data can provide robust predictions when accompanied by efficient ML models and feature selection strategies like mRMR or RFE.

Objectives of the Study

- Design and create machine learning algorithms to predict Alzheimer's Disease using structured clinical information.
- Evaluating and comparing Random Forest, Logistic Regression, and SVM classifiers
- Use Mutual Information and Recursive Feature Elimination for the purposes of selecting the features.
- Use SHAP to interpret model predictions
- Create a cost-effective and scalable early clinical diagnosis predictive framework.

Data Collection

The database used was OASIS-3, which contains 150 subjects aged 60–96 with clinical and demographic information. Cognitive measures are MMSE, CDRSUM, and CDRTOT; demographic measures are age, gender, SES, and education. Neuroimaging measures have not been included in order to keep it deployable in low-resource environments. Dataset magenta`https://www.oasis-brains.org/`.

Exploratory Data Analysis and Hypotheses

A. Diagnosis Class Distribution

The Diagnosis Class Distribution plot illustrates the distribution of subjects by diagnosis labels in the data. The plot immediately indicates that most subjects are classified as "Normal," and fewer subjects have Alzheimer's Disease and Mild Cognitive Impairment (MCI). This observable class imbalance illustrates why skewed data should be accounted for in the training of the model, which is the reason that a balanced class-weight approach was used in the algorithms to avoid bias towards the most prevalent "Normal" class

Images/Figure 1.png

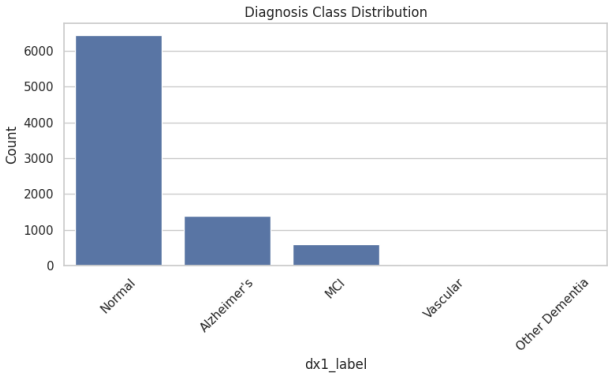

Fig. 1. Diagnosis class distribution showing the number of observations across each diagnostic category.

Images/Figure 3.png

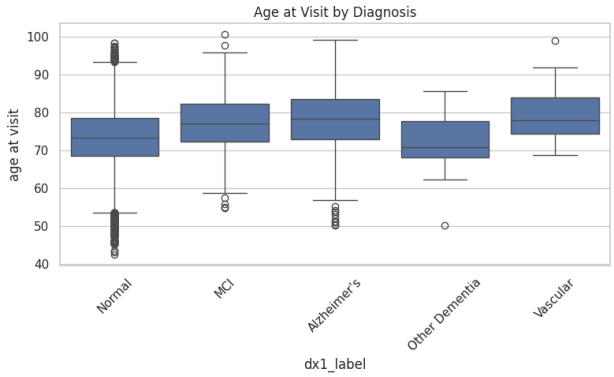

Fig. 3. Boxplot showing variation in age at visit across diagnosis categories.

B. Cognitive Stage Classification

Cognitive staging was determined through MMSE and CDRTOT threshold values to divide subjects into Healthy, Mild Cognitive Impairment (MCI), and stages of Alzheimer's Disease. Classification allows finer analysis of performance and evolution of cognition. The graphed distribution indicates the greatest proportion of subjects in the Healthy category and fewer in the moderate or severe stages to justify the use of balanced learning strategies in downstream modeling.

Images/Figure 2.png

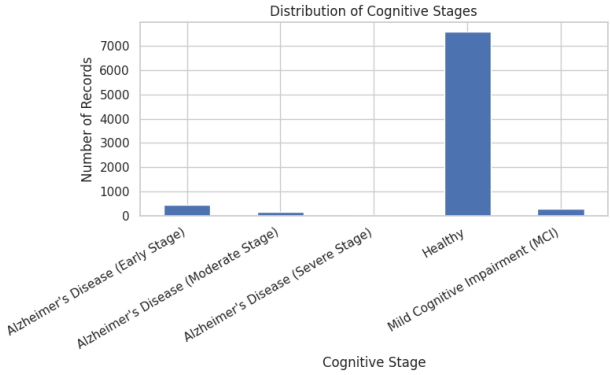

Fig. 2. Distribution of cognitive stages derived from MMSE and CDRTOT scores.

C. Age Distribution by Diagnosis

Age is a major risk factor for the onset of Alzheimer's Disease. The distribution of age by diagnosis, plotted by way of boxplots, reveals that those diagnosed as having Alzheimer's or other dementias are generally older on average than those diagnosed as Normal or having MCI. This fits with the clinical experience that increasing age contributes to susceptibility to neurodegenerative decline.

D. MMSE Score Distribution by Diagnosis

There is a clear cognitive gradient in the distribution of MMSE scores between diagnosis groups. Normal diagnoses have higher MMSE scores to represent higher cognitive performance levels, and MCI and Alzheimer's diagnoses have progressively lower and more varied scores. This validates MMSE as a helpful screening device for cognitive impairment.

Images/Figure 4.png

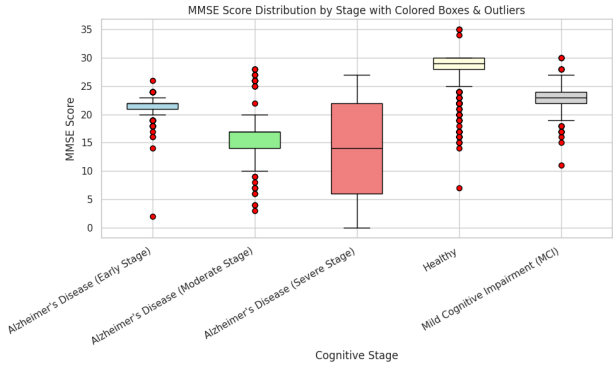

Fig. 4. Violin plot of MMSE scores by diagnosis.

E. Correlation Analysis

The heat map plot reveals correlations among the significant quantitative features, including MMSE, CDRSUM, and memory properties. A very interesting point of note is the high negative correlation between MMSE and CDRSUM that reveals that as cognitive task performance declines, the ratings for dementia increase. Moderate correlations among the subdomains of the memory also establish their significance in diagnosis.

Images/Figure 5.png

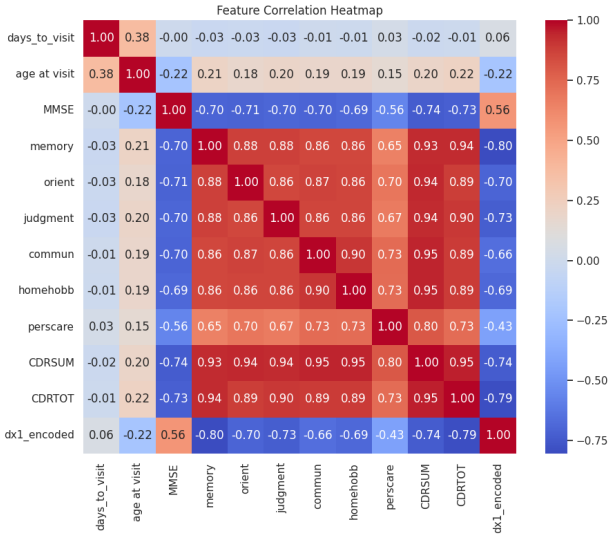

Fig. 5. Correlation heatmap showing inverse relation between MMSE and CDRSUM.

The scatter plot displays an inverse relationship between MMSE and CDRSUM wherein lower MMSE scores are associated with higher CDRSUM values. Such a trend holds across diagnostic category and graphically confirms the high negative correlation between the two indicators of decline in cognitive function..

Images/Figure 6.png

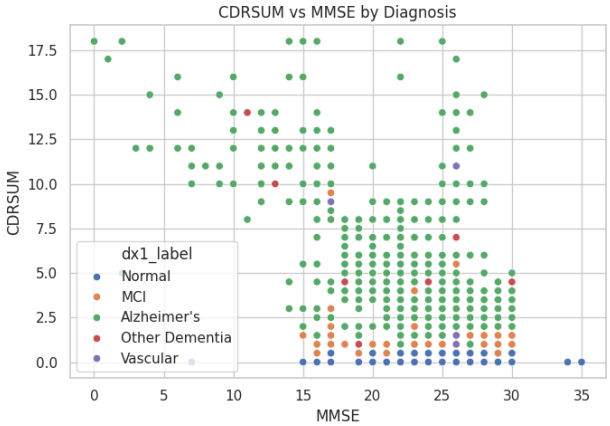

Fig. 6. Scatterplot showing the inverse relationship between MMSE and CDRSUM.

F. MMSE Trends with Age and Stage

Analysis of the MMSE score in terms of both cognitive stage and age reveals typical patterns through the disease continuum. Healthy individuals consistently score higher on the MMSE in all ages, and the moderate and severe Alzheimer's cases drop sharply. The pattern mirrors the accumulated effects of the burden of the disease and aging on cognition.

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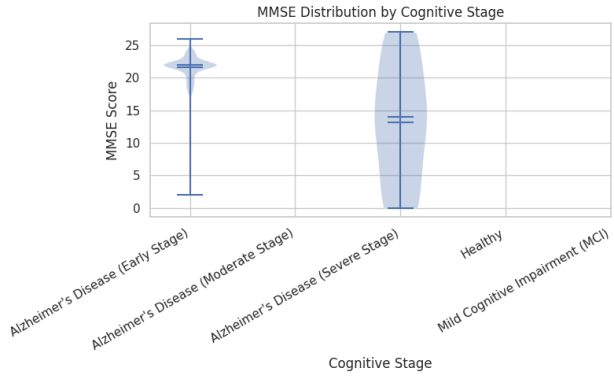

Fig. 7. MMSE score plotted against age across cognitive stages.

The MMSE and CDRSUM scores averaged across cognitive stages indicate a distinct decline in MMSE and increase in CDRSUM as the disease becomes more severe. This solidifies the negative relationship between cognitive function and rating for dementia and summarizes cognitive decline stage-wise

Images/Figure 8.png

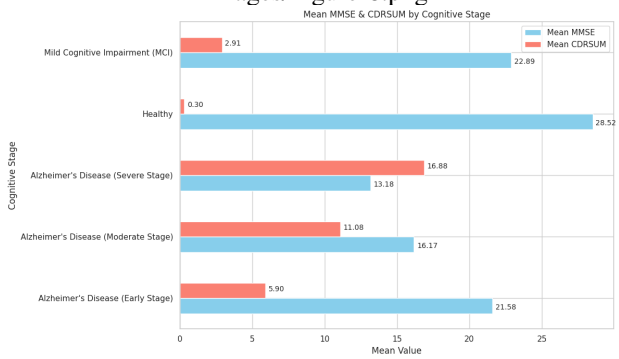

Fig. 8. Average MMSE and CDRSUM scores by cognitive stage.

Hypotheses:

H_0: ML models can accurately predict Alzheimer's using clinical features.

H_1: ML models cannot accurately predict Alzheimer's using clinical features.

Data Analytics

Images/Figure 9.png

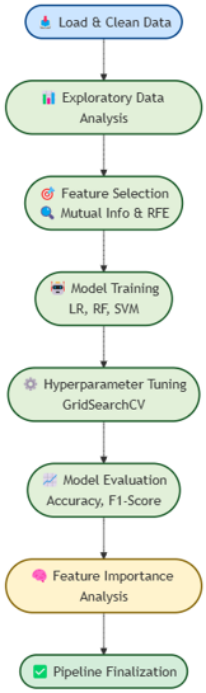

Fig. 9. ML pipeline showing preprocessing, selection, training, and evaluation.

Missing values in EDUC and SES were imputed; columns with more than 50% missing were dropped. Label encoding and StandardScaler were used. Mutual Information and RFE identified MMSE, CDRSUM, Age, EDUC, and SES as final features.

Models Trained:

- **Random Forest:** Most accurate and interpretable.
- **Logistic Regression:** Baseline, easy to interpret.
- **SVM (RBF Kernel):** Good non-linear separator.

Modeling and Evaluation

Random Forest, Logistic Regression, and Support Vector Machine (SVM) models were trained. For maximizing the relevance of features as well as the performance of the models, two most important feature selection methods were used: Recursive Feature Elimination (RFE) and Mutual Information (MI). 5-fold cross-validation was utilized to train the models and hyperparameters were tuned through GridSearchCV.

A. Feature Selection

- **Recursive Feature Elimination (RFE):** Selected top 11 features using model-estimated importance scores.
- **Mutual Information (MI):** Ranked features based on their non-linear mutual dependence with the target variable.

Top features included MMSE, CDRSUM, Age, Judgment, and HomeHobb.

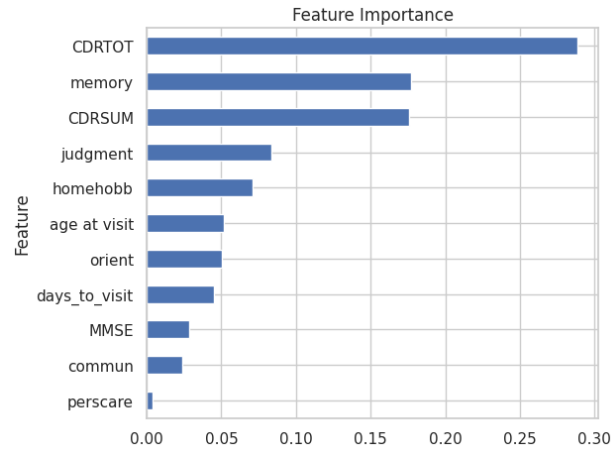

Fig. 10. Important features which are crucial for predictions in order of their importance

B. Model Configurations and Hyperparameters

- **Random Forest:** 200 estimators, max depth grid-searched. Performed best overall.
- **Logistic Regression:** L2-regularized with tuned C values. Baseline model.
- **SVM (Linear Kernel):** Tuned C parameter, used for both RFE and MI-based feature sets.

3a: Run GridSearch with RFE

```
[ ]
print("\n=== Model Comparison using RFE ===")
for name, model in models.items():
    rfe = RFE(estimator=model, n_features_to_select=11)
    X_train_rfe = rfe.fit_transform(X_train, y_train)
    X_test_rfe = rfe.transform(X_test)

    pipe = Pipeline([("clf", model)])
    grid = GridSearchCV(pipe, param_grid=param_grids[name], cv=cv, scoring='accuracy', n_jobs=-1)
    grid.fit(X_train_rfe, y_train)

    test_score = accuracy_score(y_test, grid.predict(X_test_rfe))
    print(f"{name} Accuracy: {test_score:.4f}")

    if test_score > best_score:
        best_score = test_score
        best_model = grid.best_estimator_
        best_model_name = name
        best_rfe = rfe
        X_test_best = X_test_rfe
```

```
{}
=== Model Comparison using RFE ===
Random Forest Accuracy: 0.9623
Logistic Regression Accuracy: 0.9617
SVM Accuracy: 0.9629
```

Fig. 11. Hyperparameter tuning using GridSearch

C. Model Performance with RFE

- **Random Forest:** Accuracy = 0.964; showed strong balance across all classes.
- **Logistic Regression:** Accuracy = 0.96170; performed well but slightly underfitted rare classes.

- **SVM:** Accuracy = 0.962988; captured patterns reasonably well with linear separation.

D. Model Performance with MI

- **Random Forest:** Accuracy = 0.962; consistently top performer.
- **Logistic Regression:** Accuracy = 0.961788; similar to-lightly lower than RFE.
- **SVM:** Accuracy = 0.9629.86; performed comparably but lacked flexibility for non-linear patterns.

E. Classification Report Summary

F1-scores, recall, and precision for all the models were inspected. Random Forest produced the best F1-scores for all the 5 diagnostic classes. Logistic Regression performed poorly in the recall for the minority classes such as Vascular Dementia, and SVM performed a little poorer in Alzheimer's vs MCI boundary. This validates Random Forest as a robust and interpretable model.

F. Best Model Selection

Random Forest along with RFE was chosen as the ultimate model. It performed the best in terms of test accuracy and stable performance on all the classes. The model was also examined using feature importance visualisation.

G. Test Case

Here's the performance of the model on a test case example. The model made a correct prediction as to the type and stage of Alzheimer's Disease. It accepts the chosen features upon which we trained the model and employs these to make the prediction..

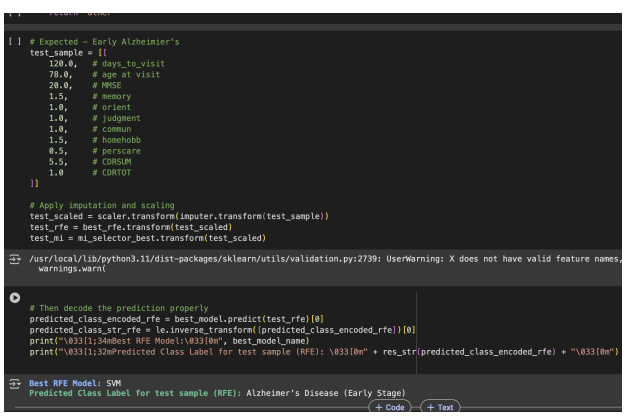

Fig. 12. Test case with inputs which is predicted to be Alzheimer's Early Stage by our selected model which is SVM

Results and Visualization

Model	Accuracy
Random Forest	96%
Logistic Regression	96.17%
SVM	96.29%

TABLE I
MODEL PERFORMANCE COMPARISON

SHAP plots confirmed MMSE and CDRSUM as top predictors. High MMSE reduced AD likelihood, while high CDRSUM increased it, aligning with clinical knowledge.

Conclusion

This work illustrates that structured cognitive and clinical data may be utilized in the construction of precise and transparent models for the prediction of Alzheimer's Disease. In the tested models, the Support Vector Machine (SVM), tuned using GridSearchCV, returned the highest accuracy rate of 96.29% which was slightly better than Random Forest and Logistic Regression. The pipeline as a whole scales and does not depend on the use of expensive imaging technologies and is therefore applicable to real-world screening in a clinical setting. Extending the model to the larger OASIS-3 database, multi-class AD staging, incorporating extra imaging markers, and deployment in electronic medical records systems for use in the clinic are areas that will be investigated in the future.

Future Scope

In future versions of this work, the pipeline can also be enriched by performing data imputation and feature scaling after the **train-test split** to preclude the leakage of the transformation acquired on the train set to the test set and consequently the impairment of the model's generalization capacity. Further, we identified notable outliers in the MMSE score distribution that may affect the model's training. For that reason, more sophisticated outlier detection algorithms like Isolation Forest or robust scalers (like RobustScaler in sklearn) may be used in a way that does not reduce the data distribution but lessens the impact of outlier values. Other advancements include the addition of longitudinal MRI scans to the OASIS-3 database to capture the disease's evolution through the evolution of time and the application of deep learning algorithms like LSTMs or 3D CNNs to temporal and spatial pattern identification. Finally, the application of the model as a real-time SHAP-based interpretability clinical decision support system will provide more ease of use to practitioners to enable evidence-based risk appraisals.

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